

EXP 1) Regular Expression Pattern Matching

CODE

```
import re

text = input("Enter a sentence: ")

phone = re.findall(r"\d{10}", text)

email = re.findall(r"\S+@\S+", text)

print("Phone Number:", phone)

print("Email:", email)

word = input("Enter the word to search: ")

if re.search(word, text):

    print("Word Found")

else:

    print("Word Not Found")
```

Output:

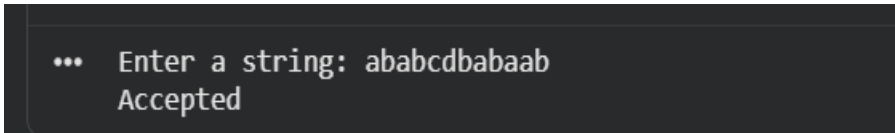
```
... Enter a sentence: my name is nischay and my mail is niown@gmail.com 944555
Phone Number: ['9445555268']
Email: ['niown@gmail.com']
Enter the word to search: hjiis
Word Not Found
```

2) Finite State Automaton (FSA) for String Pattern Recognition

CODE:

```
def finite_state_automaton(string):  
    return string.endswith("ab")  
  
s = input("Enter a string: ")  
  
if finite_state_automaton(s):  
    print("Accepted")  
else:  
    print("Rejected")
```

OUTPUT:



```
... Enter a string: ababcdbabaab  
Accepted
```

3) Morphological Analysis Using NLTK

CODE:

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.stem import WordNetLemmatizer

nltk.download('punkt')

nltk.download('wordnet')

nltk.download('omw-1.4')

text = input("Enter a sentence: ")

words = word_tokenize(text)

lemmatizer = WordNetLemmatizer()

print("Word\tLemma")

for word in words:

    print(word, "\t", lemmatizer.lemmatize(word))
```

OUTPUT:

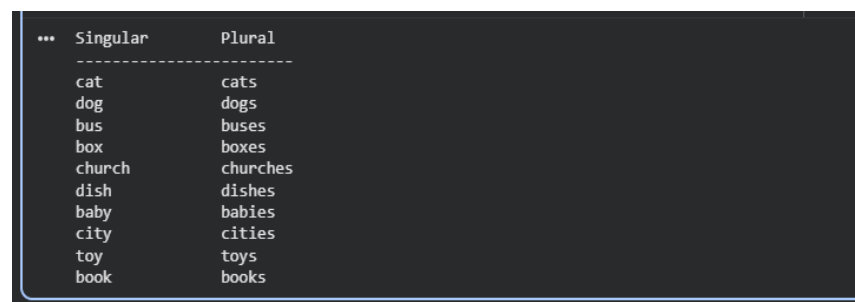
```
... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
Enter a sentence: Boys are playing games
Word      Lemma
Boys      Boys
are       are
playing   playing
games     game
```

4) Finite State Machine for Morphological Parsing

CODE:

```
def generate_plural(noun):  
    if noun.endswith(("s", "x", "z", "ch", "sh")):  
        plural = noun + "es"  
    elif noun.endswith("y") and len(noun) > 1 and noun[-2].lower() not in "aeiou":  
        plural = noun[:-1] + "ies"  
    else:  
        plural = noun + "s"  
    return plural  
  
words = ["cat", "dog", "bus", "box", "church",  
        "dish", "baby", "city", "toy", "book"]  
print("Singular\tPlural")  
print("-----")  
for word in words:  
    print(f'{word:10}\t{generate_plural(word)}')
```

OUTPUT:



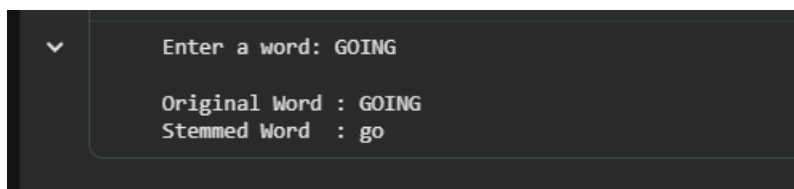
```
*** Singular      Plural  
-----  
cat              cats  
dog              dogs  
bus              buses  
box              boxes  
church           churches  
dish             dishes  
baby             babies  
city             cities  
toy              toys  
book             books
```

5) Word Stemming Using the Porter Stemmer Algorithm

CODE

```
import nltk  
  
from nltk.stem import PorterStemmer  
  
stemmer = PorterStemmer()  
  
word = input("Enter a word: ")  
  
stem = stemmer.stem(word)  
  
# Display result  
  
print("\nOriginal Word :", word)  
print("Stemmed Word :", stem)
```

OUTPUT:

A screenshot of a terminal window with a dark background. On the left side, there is a small white downward-pointing arrow icon. The terminal displays the following text: "Enter a word: GOING", "Original Word : GOING", and "Stemmed Word : go".

```
Enter a word: GOING  
  
Original Word : GOING  
Stemmed Word : go
```

6) Bigram (N-Gram) Model for Text Generation

CODE:

```
import nltk
from nltk import bigrams
from collections import defaultdict
import random

nltk.download('punkt')
nltk.download('punkt_tab')

text = input("Enter a sentence: ")
words = nltk.word_tokenize(text.lower())
model = defaultdict(list)

for w1, w2 in bigrams(words):
    model[w1].append(w2)

start = input("Enter the starting word: ").lower()
if start not in model:
    print("Starting word not found in the sentence.")
else:
    generated = [start]
    current = start

    for i in range(10):
        if current not in model:
            break

        current = random.choice(model[current])
        generated.append(current)

    print("\nGenerated Text:")
    print(" ".join(generated))
```

OUTPUT:

```
▼ ... [nltk_data] Downloading package punkt to /root/nltk_data...  
[nltk_data] Package punkt is already up-to-date!  
[nltk_data] Downloading package punkt_tab to /root/nltk_data...  
[nltk_data] Package punkt_tab is already up-to-date!  
Enter a sentence: MY NAME IS NISCHAY  
Enter the starting word: MY  
  
Generated Text:  
my name is nischay
```